3. Eigenvectors, eigenvalues, Spectral theorem

(a) We know it holds: $A = T\Lambda T^{-1}$

Thus implies that : $AT = T\Lambda T^{-1}T = T\Lambda \Rightarrow AT = T\Lambda$

From the matrix product, it holds:

$$AT = A \cdot \begin{bmatrix} | & | & & | \\ t^{(1)} & t^{(2)} & \cdots & t^{(n)} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ At^{(1)} & At^{(2)} & \cdots & At^{(n)} \\ | & | & & | \end{bmatrix}$$

On the other hand

$$T\Lambda = \begin{bmatrix} | & | & & | \\ t^{(1)} & t^{(2)} & \cdots & t^{(n)} \\ | & | & & | \end{bmatrix} \begin{bmatrix} - \Lambda^{(1)} - \\ - \Lambda^{(2)} - \\ \vdots \\ - \Lambda^{(n)} - \end{bmatrix} = t^{(1)}\Lambda^{(1)} + t^{(2)}\Lambda^{(2)} + \dots + t^{(n)}\Lambda^{(n)}$$

which is a sum of $n \times n$ matrices.

Let's expand the first of these matrices, to understand its structure.

$$t^{(1)}\Lambda^{(1)} = \begin{bmatrix} t_1^{(1)} \\ t_2^{(1)} \\ \vdots \\ t_n^{(1)} \end{bmatrix} \begin{bmatrix} \lambda_1 & \bar{0} \end{bmatrix} = \begin{bmatrix} \lambda_1 t^{(1)} & \bar{0} \end{bmatrix}$$

Therefore, summing the n matrices yields:

$$T\Lambda = \begin{bmatrix} | & | & & | \\ \lambda_1 t^{(1)} & \lambda_2 t^{(2)} & \cdots & \lambda_n t^{(n)} \\ | & | & & | \end{bmatrix}$$

If two matrices are equal, so are their columns. From this, it follows:

$$\begin{bmatrix} | & | & & | \\ At^{(1)} & At^{(2)} & \cdots & At^{(n)} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \lambda_1 t^{(1)} & \lambda_2 t^{(2)} & \cdots & \lambda_n t^{(n)} \\ | & | & & | \end{bmatrix}$$

Therefore $At^{(i)} = \lambda_i t^{(i)}$

(b) We ~~also~~ know $A = U\Lambda U^T$, which implies $AU = U\Lambda U^T U = U\Lambda I$ because U is orthogonal and $U^T U = I$ for such matrices

Therefore $AU = U\Lambda$. Following a similar demonstration as above:

$$AU = \begin{bmatrix} | & | & & | \\ Au^{(1)} & Au^{(2)} & \cdots & Au^{(n)} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \lambda_1 u^{(1)} & \lambda_2 u^{(2)} & \cdots & \lambda_n u^{(n)} \\ | & | & & | \end{bmatrix} = U\Lambda$$

Therefore $Au^{(i)} = \lambda_i u^{(i)}$

(c) If A is PSD, it holds: $A^T = A$, $x^T A x \geq 0 \quad \forall x \in \mathbb{R}^n$

Since A is symmetric, it can also be re-written as $A = U\Lambda U^T$

We can define $\Lambda^{1/2}$ such that $\Lambda^{1/2} \cdot \Lambda^{1/2} = \Lambda$. Since Λ is diagonal, so is $\Lambda^{1/2}$.

Therefore $\Lambda^{1/2}$ is also symmetric.

It follows:

$$x^T A x \geq 0 \Rightarrow x^T U \Lambda^{1/2} \Lambda^{1/2} U^T x \geq 0 \quad \forall x \in \mathbb{R}^n$$

$$(\Lambda^{1/2} U^T x)^T (\Lambda^{1/2} U^T x) \geq 0$$

Let's call $\hat{x} = U^T x$. It follows $\sum_{i=1}^{n} \lambda_i \hat{x}_i^2 \geq 0 \quad \forall x \in \mathbb{R}^n$

which is satisfied for all x in $\mathbb{R}^n$ if and only if $\lambda_i \geq 0 \quad \forall i \in [1, n]$

Solutions show a different derivation, checking $(t^{(i)})^T A t^{(i)} \geq 0$